

BCNF Normalization

1. Users(userId: INT [PK], username: VARCHAR(30) UNIQUE NOT NULL, password: VARCHAR(30), firstName: VARCHAR(30), lastName VARCHAR(30))

Users BCNF:

(superkey) userId -> username, password, firstName, lastName

2. Sets(setId: INT [PK], userId: INT [FK to ExerciseLog.userId], exercisId: INT [FK to ExerciseLog.exercisId], order: INT, lbs: REAL, reps: INT)

Sets BCNF:

(superkey) setId -> userId, exercisId, order, lbs, reps

3. WorkoutTemplates(workoutId: INT [PK], userId: INT [FK to Users.userId], lastDate: DATE, name: VARCHAR(255))

WorkoutTemplates BCNF:

(superkey) workoutId -> userId, lastDate, name

4. ExerciseLog(userId: INT [PK, FK to Users.userId], exercisId: INT [PK, FK to Exercises.exercisId], like: BOOLEAN, timesCompleted: INT)

ExerciseLog BCNF:

(superkey) (userId, exercisId) -> userId, like, timesCompleted

(superkey) (userId, exercisId) -> exercisId, like, timesCompleted

5. Exercises(exercisId: INT [PK], name: VARCHAR(255), desc: VARCHAR(500), video: VARCHAR(255), ownerId: INT [FK to Users.userId])

Exercises BCNF:

(superkey) exercisId -> name, desc, video, ownerId

6. WorkoutContents(workoutId: INT [PK, FK to WorkoutTemplates.workoutId], exercisId: INT [PK, FK to Exercises.exercisId], order: INT)

WorkoutContents BCNF:

(superkey) (workoutId, exercisId) -> workoutId, order

(superkey) (workoutId, exercisId) -> exercisId, order

7. Muscles(muscleId: INT [PK], name: VARCHAR(255) UNIQUE)

Muscles BCNF:

(superkey) muscleId -> name

ExercisesMuscles BCNF:

8. ExercisesMuscles(exerciseld: INT [PK, FK to Exercises.exerciseld], musculeld: INT [PK, FK to Muscles.musculeld])

ExercisesMuscles BCNF:

(superkey) (exerciseld, musculeld) -> musculeld

(superkey) (exerciseld, musculeld) -> exerciseld

Full Normalized Logical Design

Users(userid: INT [PK], username: VARCHAR(30) UNIQUE NOT NULL, password: VARCHAR(30), firstName: VARCHAR(30), lastName VARCHAR(30))

Sets(setId: INT [PK], userId: INT [FK to ExerciseLog.userId], exerciseld: INT [FK to ExerciseLog.exerciseld], order: INT, lbs: REAL, reps: INT)

WorkoutTemplates(workoutId: INT [PK], userId: INT [FK to Users.userId], lastDate: DATE, name: VARCHAR(255))

ExerciseLog(userId: INT [PK, FK to Users.userId], exerciseld: INT [PK, FK to Exercises.exerciseld], like: BOOLEAN, timesCompleted: INT)

Exercises(exerciseld: INT [PK], name: VARCHAR(255), desc: VARCHAR(500), video: VARCHAR(255), ownerId: INT [FK to Users.userId])

WorkoutContents(workoutId: INT [PK, FK to WorkoutTemplates.workoutId], exerciseld: INT [PK, FK to Exercises.exerciseld], order: INT)

Muscles(musculeld: INT [PK], name: VARCHAR(255) UNIQUE)

ExercisesMuscles(exerciseld: INT [PK, FK to Exercises.exerciseld], musculeld: INT [PK, FK to Muscles.musculeld])